

$^{32}\text{S}(^3\text{He},n)$ 1986A115

$^{32}\text{S}+2p \rightarrow ^{34}\text{Ar}$ from $J^\pi=0^+$ ^{32}S ground state.

1986A115: a 25.4-MeV ^3He beam was produced from the time-of-flight facility at the University of Colorado. The target was natural sulfur evaporated onto a 2-mg/cm² gold foil. Neutrons were detected using three NE224 liquid scintillators with FWHM=350 keV. Measured $\sigma(E_n, \theta)$. Deduced levels, L-transfers, J, and π from local, zero-range DWUCK-DWBA analysis. Comparisons with shell model and analog states in ^{34}S .

1972Bb01: 14-MeV ^3He beam from the 7 MV Van de Graaff accelerator at Frankfurt University impinged on natural ^{32}S targets. Measured $\sigma(E_n, \theta)$. Deduced ground state, J, and π from zero-range DWUCK-DWBA analysis.

1968Ha09: 4.5-6.2-MeV ^3He beams from the de Graaff accelerator at Hahn-Meitner-Institut für Kernforschung Berlin and impinged on FeS targets. Measured neutron spectra using a time-of-flight spectrometer. Deduced 5 levels.

1967Mi02: 10-12-MeV ^3He beams from the ONR-CIT tandem accelerator impinged on natural sulfur targets. Neutron energies were measured by means of (n, α) reactions induced by neutrons in the active volume of a silicon semiconductor detector. Deduced Q value and one level.

1967Mc03: 4.9-5.6-MeV ^3He beams from the SUNI 6 MeV Van de Graaff accelerator at Southern Universities Nuclear Institute, South Africa impinged on natural CdS targets. Neutrons were detected using a NE213 liquid scintillator. Measured $\sigma(E_n, \theta)$ and deduced g.s. and 2 levels.

1984Be05: 4.6-6.5 MeV ^3He beams from the University of Kentucky Van de Graaff Accelerator impinged on CdS and ZnS targets. Neutron angular distributions and excitation curves were measured with a neutron time-of-flight facility.

 ^{34}Ar Levels

ε : enhancement factor = $(d\sigma/d\omega)_{\text{exp}}/[213(d\sigma/d\omega)_{\text{DWBA}}]$.

E(level) [†]	L [†]	Comments
0	0	$\varepsilon=1.4$.
2103 30	2	$\sigma(\text{exp})/\sigma(\text{theory})=1.84$ or 1.68 (1972Bb01). E(level): weighted average of 2090 30 (1986A115), 2100 30 (1968Ha09), 2190 40 (1967Mc03), and 2058 35 (1967Mi02).
3295 30	(2)	$\varepsilon=3.9$. E(level): weighted average of 3290 30 (1986A115) and 3300 30 (1968Ha09).
3900 30	(0)	$\varepsilon=0.32$. E(level): weighted average of 3900 70 (1986A115) and 3900 30 (1968Ha09). Other: 3590 60 (1967Mc03), a broad peak, possibly unresolved 3295+3900.
4050 [‡] 30		$\varepsilon=6.1$.
4150 [‡] 30		
4510 30	3	$\varepsilon=2.9$.
4950 50	0	$\varepsilon=25$.
5310 30	(5)	$\varepsilon=0.38$.
5620 30	2	$\varepsilon=18$.
5930 50	0	
6470 30		
6820 40		
6990 50		
7300 30		

[†] From **1986A115**, unless otherwise noted.

[‡] From **1968Ha09**.